AYMANE AIT DADS

Machine Learning Engineer Intern | Neural Decoding · Time Series · Deep Learning · BCI

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PROFESSIONAL SUMMARY

Ranked 17th out of 3,000+ teams in the NVIDIA Nemotron Reasoning Challenge (score 0.86/1.0, $106K+ prize pool) by designing and executing an end-to-end LoRA SFT pipeline on a 30B-parameter Mamba-Transformer with BF16 precision, 8,192-token context, and completion-only loss. Reduced manual network analysis time by 60% at Orange Maroc by building production-grade Random Forest and K-Means pipelines on 100K+ records, delivering insights directly to the network optimization team. Brings deep hands-on experience in designing machine learning models, analyzing complex datasets, and communicating results clearly to both technical and non-technical stakeholders - skills directly applicable to developing novel neural decoders for BCI applications. Aims to contribute rigorous ML engineering and ablation-driven model development to Neuralink's mission of restoring movement and communication for people with paralysis.

CORE COMPETENCIES

Machine Learning Models | Complex Dataset Analysis | Time Series Data | Neural Decoding | Cross-functional Collaboration | Clean Scalable Code | High-impact Projects | Communicating Results

WORK EXPERIENCE

Orange Maroc
Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Built and evaluated Random Forest classification and K-Means clustering models on **100K+ fiber-optic network time series records** (upload speed, latency, jitter) to classify network quality across thousands of nodes.
- Designed an end-to-end clustering pipeline mapping customer performance profiles to **5 commercial service tiers**; output adopted directly by the network optimization team for infrastructure decisions.
- Reduced **manual analysis time by ~60%** through automated feature engineering, EDA, and model evaluation workflows built entirely in Python, Pandas, and Scikit-learn.
- Delivered stakeholder presentations translating complex model outputs into **actionable maintenance recommendations**, communicating results clearly to both technical engineers and non-technical business stakeholders.

PROJECTS

NVIDIA Nemotron Model Reasoning Challenge
Kaggle Competition · In Progress (Jun 2026) · 17th / 3,000+ Teams

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, $106K+ prize pool) in the ongoing NVIDIA Nemotron Reasoning Challenge by fine-tuning a **30B-parameter Mamba-Transformer** with LoRA SFT (r=32, BF16, completion-only loss, 8,192-token context, grad_accum=32), owning the full training pipeline end-to-end.
- Ran **controlled ablation experiments** across learning rate schedules, packing strategies, and synthetic augmentation - discovering that reasoning-style consistency outweighs data volume in LLM SFT; built custom cipher parsers and an 8-bit operation solver to produce 669 fully verified chain-of-thought training rows.

PyTorch · Unsloth · TRL · PEFT · Hugging Face Transformers · vLLM · Kaggle GPU

Anomalous Sound Detection - Industrial Equipment
EURECOM · 2025 · AUC 0.80

- Designed an unsupervised fault-detection system for slide-rail machinery using a Transformer autoencoder on **Mel spectrograms with SpecAugment**, achieving AUC 0.80 on unlabeled time series audio data with no anomaly labels available during training.
- Applied signal preprocessing and spectrogram feature engineering to convert raw **time series sensor streams** into structured inputs suitable for deep learning, directly mirroring the unstructured data processing challenges present in neural signal decoding.

PyTorch · Mel Spectrograms · SpecAugment · Transformer Autoencoder

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM
Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Statistics, Computer Vision, NLP, Image Security, Cloud Computing

CPGE - Mathematics & Physics - Ibn Timiya
2021 - 2023

SKILLS

ML Frameworks & Engineering: PyTorch | Scikit-learn | Hugging Face Transformers | Unsloth | TRL | PEFT | vLLM

ML Methods & Techniques: LoRA SFT | BF16/FP16 Mixed Precision | Ablation Studies | Time Series Analysis | Ensemble Methods | Clustering | Feature Engineering

Data & Infrastructure: Python | Pandas | NumPy | SQL | Git | Linux | Power BI | Docker